

1. Administrative & Study Identification

Full Study Title: Efficacy and Safety of Pembrolizumab (MK-3475) Plus Enzalutamide Plus Androgen Deprivation Therapy (ADT) Versus Placebo Plus Enzalutamide Plus ADT in Participants With Metastatic Hormone-Sensitive Prostate Cancer (mHSPC) (MK-3475-991/KEYNOTE-991)-China Extension **Short Title/Acronym:** KEYNOTE-991 China Extension **Protocol Number:** MK-3475-991 **NCT Number:** NCT04934722 **Sponsor Name:** Merck / Merck Sharp & Dohme LLC **Investigational Product:** Pembrolizumab (MK-3475) + Enzalutamide (Trade Name: Xtandi, ATC Code: L02BB04) **Phase of Development:** Phase III **Trial Status:** Completed **Medical Monitor:** Clinical Operations Lead, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (Radiographic Progression-Free Survival [rPFS] and Overall Survival [OS]) and safety of pembrolizumab plus enzalutamide plus ADT versus placebo plus enzalutamide plus ADT in participants with metastatic hormone-sensitive prostate cancer (mHSPC).

Secondary Objectives: To evaluate the Time to Initiation of the First Subsequent Anti-cancer Therapy or Death (TFST), Time to First Symptomatic Skeletal-related Event (TTSSRE), and Time to Prostate-specific Antigen (PSA) Progression.

2.2 Background & Rationale To address the gap in treatment for patients with metastatic castration-sensitive/hormone-sensitive prostate cancer (mCSPC/mHSPC). Despite standard initial androgen deprivation therapy (ADT) and the addition of potent androgen receptor signaling inhibitors (ARSIs) like enzalutamide, patients inevitably progress to castration-resistant disease. The addition of an immune checkpoint inhibitor (pembrolizumab, targeting PD-1) aims to harness the inflammatory response generated by ADT-induced apoptosis, potentially inducing a synergistic anti-tumor immune response and delaying disease progression.

Note: As of Amendment 4, the study was stopped for futility. All study participants were instructed to stop ongoing treatment with pembrolizumab/placebo, barring exceptions for participants who, in the assessment of their study physician, were benefiting from the combination after consulting with the Sponsor.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Active Comparator (Enzalutamide + ADT)
Blinding: Double-blind **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target \$N\$: 186 participants (for the China Extension cohort) **Number of Sites:** Multicenter (China specific) **Estimated Study Start:** July 2021 **Estimated Primary Completion:** Completed (Study stopped for futility)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants with histologically- or cytologically-confirmed adenocarcinoma of the prostate without small cell histology. 2. Has metastatic disease assessed by investigator and initially verified by Blinded Independent Central Review (BICR) by either ≥ 2 bone lesions on bone scan and/or visceral disease by computed tomography/magnetic resonance imaging (CT/MRI). 3. Willing to maintain continuous Androgen Deprivation Therapy (ADT) with a luteinizing-hormone releasing hormone (LHRH) agonist or antagonist during study treatment or have a history of bilateral orchiectomy. 4. Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 assessed within 10 days of randomization. 5. Participants receiving bone resorptive therapy (e.g., bisphosphonate or denosumab) must have been on stable doses prior to randomization. 6. Has adequate organ function and has provided newly obtained core or excisional biopsy (within 12 months of screening) from soft tissue not previously irradiated (or bone biopsy for bone-only/predominant disease). 7. Male participants must agree to refrain from donating sperm and use contraception/remain abstinent during the intervention period and for at least 120 days after the last dose of study intervention.

4.2 Exclusion Criteria 1. Has a known additional malignancy that is progressing or has required active treatment in the last 3 years. 2. Has an active autoimmune disease that has required systemic treatment in the past 2 years, a diagnosis of immunodeficiency, or is receiving chronic systemic steroid therapy. 3. Prior treatment with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent, or prior treatment with a next-generation hormonal agent (e.g., abiraterone, enzalutamide, apalutamide, darolutamide). 4. Has undergone major surgery including local prostate intervention within 28 days prior to randomization. 5. Has a gastrointestinal disorder affecting absorption or is unable to swallow tablets/capsules. 6. Has an active infection (including tuberculosis) requiring systemic therapy, known active HIV, HBV, or HCV infection. 7. Has a history of (non-infectious) pneumonitis that required steroids or current pneumonitis. 8. Has known or suspected central nervous system (CNS) metastases and/or

carcinomatous meningitis. 9. Has a history of seizure, condition predisposing to seizure, or loss of consciousness within 12 months of screening. 10. Has had myocardial infarction, uncontrolled angina, severe congestive heart failure (NYHA class III/IV), clinically significant ventricular arrhythmias, severe hypotension, or uncontrolled hypertension. 11. Has a "superscan" bone scan or has had an allogenic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab 200 mg every 3 weeks (Q3W) plus Enzalutamide 160 mg once daily. **Route of Administration:** Intravenous (IV) for Pembrolizumab/Placebo; Oral for Enzalutamide. **Duration of Treatment:** Originally planned until disease progression, unacceptable toxicity, or maximum protocol-defined cycles. Due to Amendment 4 stopping the study for futility, treatment was halted for most participants. Second Course treatment is not an option.

5.2 Concomitant & Prohibited Therapy Permitted: Continuous ADT (LHRH agonists/antagonists) is required. Up to 6 previous cycles of docetaxel are allowed if completed within 2 months of randomization with no evidence of progression. **Prohibited:** Concurrent treatment with other investigational agents, other anti-cancer therapies, live vaccines within 30 days prior to randomization, or chronic systemic corticosteroids.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Imaging (CT/MRI/Bone Scan), Labs, Biopsy
Baseline	Day 0	Randomization, First Dose of IV Pembrolizumab/Placebo + Oral Enzalutamide
Interim	Every 3 Weeks	IV Dosing, Safety Labs, Efficacy Assessment (Imaging typically every 9-12 weeks)
End of Study	30 Days Post-Last Dose	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Radiographic Progression-Free Survival (rPFS): Defined as the time from randomization to occurrence of radiological tumor progression using PCWG-modified RECIST 1.1 or death due to any cause. 2. Overall Survival (OS): Time from randomization to death due to any cause. **Measurement Method:** Originally assessed by Blinded Independent Central Review (BICR). As of Amendment 04, disease progression is no longer centrally verified and is only assessed locally.

7.2 Secondary Endpoints 1. Time to Initiation of the First Subsequent Anti-cancer Therapy or Death (TFST). 2. Time to First Symptomatic Skeletal-related Event (TTSSRE). 3. Time to Prostate-specific Antigen (PSA) Progression.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring throughout the study and for 30 days post-treatment (up to 90 days for serious/immune-related AEs or 120 days for contraceptive follow-up). **Serious Adverse Events (SAEs):** Reported to the sponsor/regulatory authorities typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An external, independent committee reviews safety and efficacy data, which led to the determination of futility under Amendment 4.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy endpoints; Safety Set for all patients who receive at least one dose of study medication. **Sample Size Justification:** Target \$N = 186\$ participants (China Extension). **Handling of Missing Data / Early Termination (Amendment 4):** As the study was stopped for futility, all pre-specified interim analyses after interim analysis 1 (IA1) and final analyses described in the SAP will not be performed. Safety analysis will be performed at the end of the study; no further analyses for efficacy and electronic patient-reported outcome (ePRO) endpoints collected beyond the IA1 cutoff date will be conducted. Participants without a documented event at the time of analysis are censored at the date of last disease assessment or last follow-up.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to verify adherence to protocol and accurate data entry. **Audit**

Trail: Comprehensive eCRF locking and query resolution via standard data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-3475-991 **Registry Number:** NCT04934722 **Sponsor/Lead Organization:** Merck / Merck Sharp & Dohme LLC **Study Title:** KEYNOTE-991 China Extension **Official Regulatory Title:** A Phase 3, Randomized, Double-blind Trial of Pembrolizumab (MK-3475) Plus Enzalutamide Plus ADT Versus Placebo Plus Enzalutamide Plus ADT

in Participants With Metastatic Hormone-Sensitive Prostate Cancer (mHSPC) (KEYNOTE-991)

12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: Metastatic Hormone-Sensitive Prostate Cancer (mHSPC) / Hormone-dependent prostate cancer (SNOMED: 762883002)

Diagnosis Threshold: Histologically confirmed prostate adenocarcinoma with ≥ 2 bone lesions or visceral metastases

Medical Methodology: Androgen Deprivation Therapy (ADT) combined with Androgen Receptor Signaling Inhibitor (ARSI) and PD-1

Immunotherapy **Optimization Target:** Delaying progression to Castration-Resistant Prostate Cancer (CRPC) and improving Overall Survival

13. Study Procedures & Methodology

Management Pathway: Standard systemic therapy framework for metastatic prostate cancer; post-futility, patients are offered standard of care (SOC) treatment. **Education Protocol (HCP):** Site initiation visits and ongoing investigator training on irAE management and protocol amendments (e.g., local assessment of disease progression post-Amendment 04). **Education Protocol**

(Patient): Informed consent process and self-management guidelines for oral Enzalutamide compliance. **Quality Audit**

Mechanism: Originally utilized independent central imaging review (BICR) to ensure unbiased rPFS assessment; transitioned to local assessment via Amendment 04.

14. Observation & Follow-Up Schedule

Total Duration: Terminated early for futility; originally planned until disease progression or up to roughly 2 years (35 cycles) of immunotherapy. **Onsite Visit Cadence:** Every 3 weeks (Q3W)

corresponding to IV infusion cycles (for those remaining on exception therapy). **Remote Monitoring Frequency:** Periodic follow-up during the survival phase. **Checklist Review Interval:** Tumor imaging every 9 to 12 weeks as per protocol (locally assessed).

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				